

Fig. 3: A single-side PCB layout in actual size for seven-colour LED lighting circuit

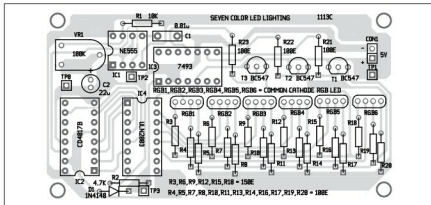


Fig. 4: Component layout for the PCB

high input clock to IC3 each time after the sixth clock of IC1.

IC3 is a four-bit binary counter, and its outputs QB through QD are connected to the bases of transistors T1 through T3 via resistors R21 through R23, respectively. The binary outputs of IC3 make corresponding transistors conduct and select a different colour after each cycle (see Table I).

The common-cathode pins of each RGB LED are connected to the outputs

of IC4 as shown in Fig. 2. ULN2003 is used for driving these LEDs. You can add even more RGB LEDs in parallel to the existing ones to make a larger lighting system.

The working of the circuit is simple. When the circuit is switched on, the clock pulse is generated from pin 3 of the timer IC1, making outputs of IC2 high, sequentially, which decides the switching rate of LEDs that can be changed through VR1. The colours are

PARTS LIST

Semiconductors:

- IC1 - NE555 timer
- IC2 - CD4017 decade counter
- IC3 - 7493 binary counter
- IC4 - ULN2003 relay driver
- T1-T3 - BC547 npn transistor
- LED1-LED6 - RGB common-cathode LED
- D1 - 1N4148 switching diode

Resistors (all $\frac{1}{4}$ -watt, $\pm 5\%$ carbon):

- R1 - 10 kilo-ohm
- R2 - 4.7 kilo-ohm
- R3, R6, R9, R12, R15, R18, R4, R5, R7, R8, R10, R11, R13, R14, R16, R17, R19, R20-R23 - 100 ohm
- VR1 - 100 kilo-ohm preset

Capacitors:

- C1 - 0.01 μ F ceramic disk
- C2 - 22 μ F, 16V electrolytic

Miscellaneous:

- CON1 - 2-pin connector
- 5V regulated DC/adaptor

selected by the binary output of IC3, which changes after every sixth pulse from IC1.

Construction and testing

A single-side PCB for seven-colour LED lighting is shown in actual size in Fig. 3, and its component layout is shown in Fig. 4. After assembling the circuit on a PCB, enclose it in a suitable case. You can wire LEDs the way you like and connect on the PCB. The circuit uses a 5V DC supply.

To test if the circuit is functioning properly, check the input supply at TP1 with respect to TP0. The output pulses generated by IC1 can be checked at TP2 with the help of an oscilloscope. The frequency of these clock pulses can be varied by VR1. The clock pulse to the binary counter IC3 can be checked at TP3. ●

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